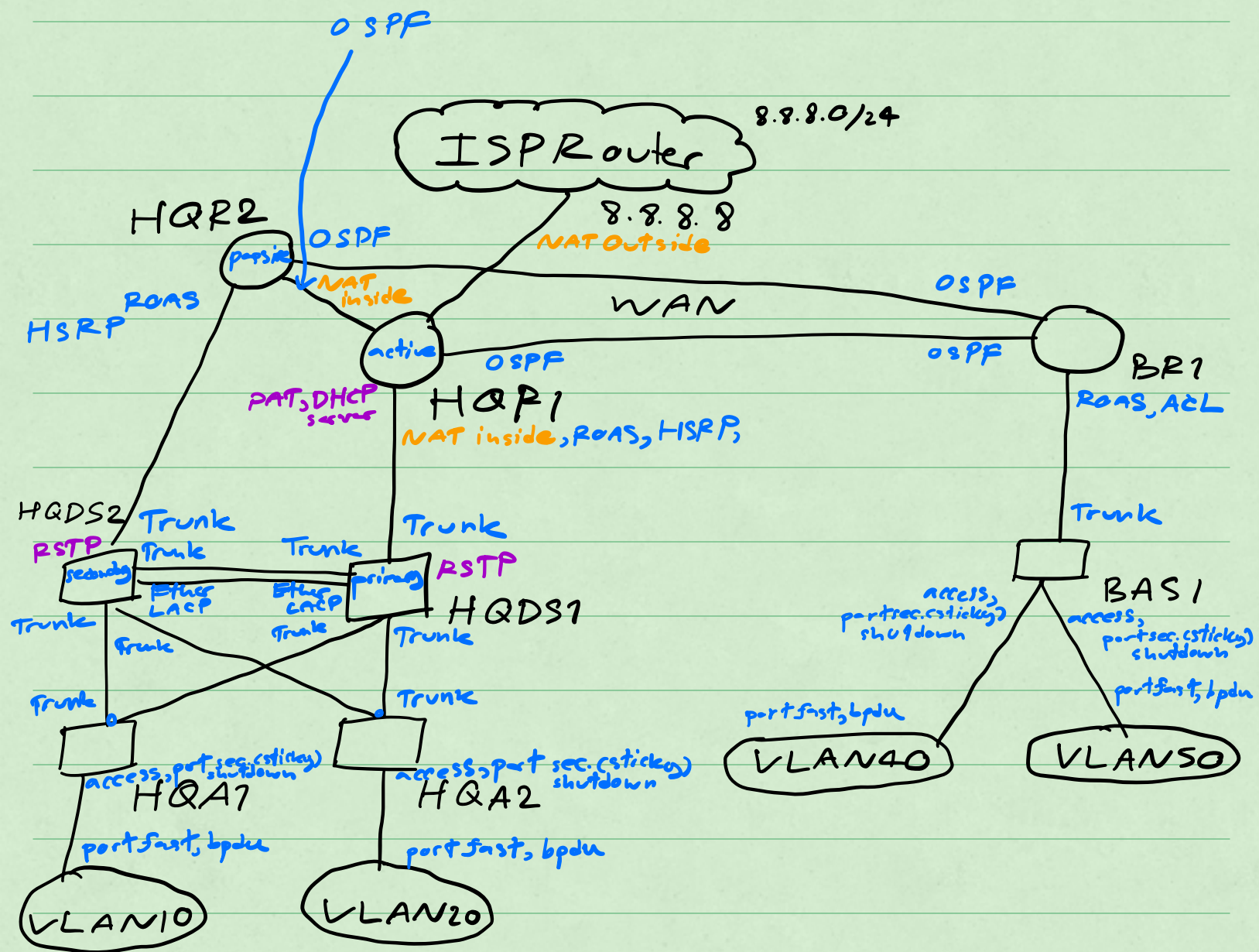




○ router

symbols → □ L2 switch

cloud / internet

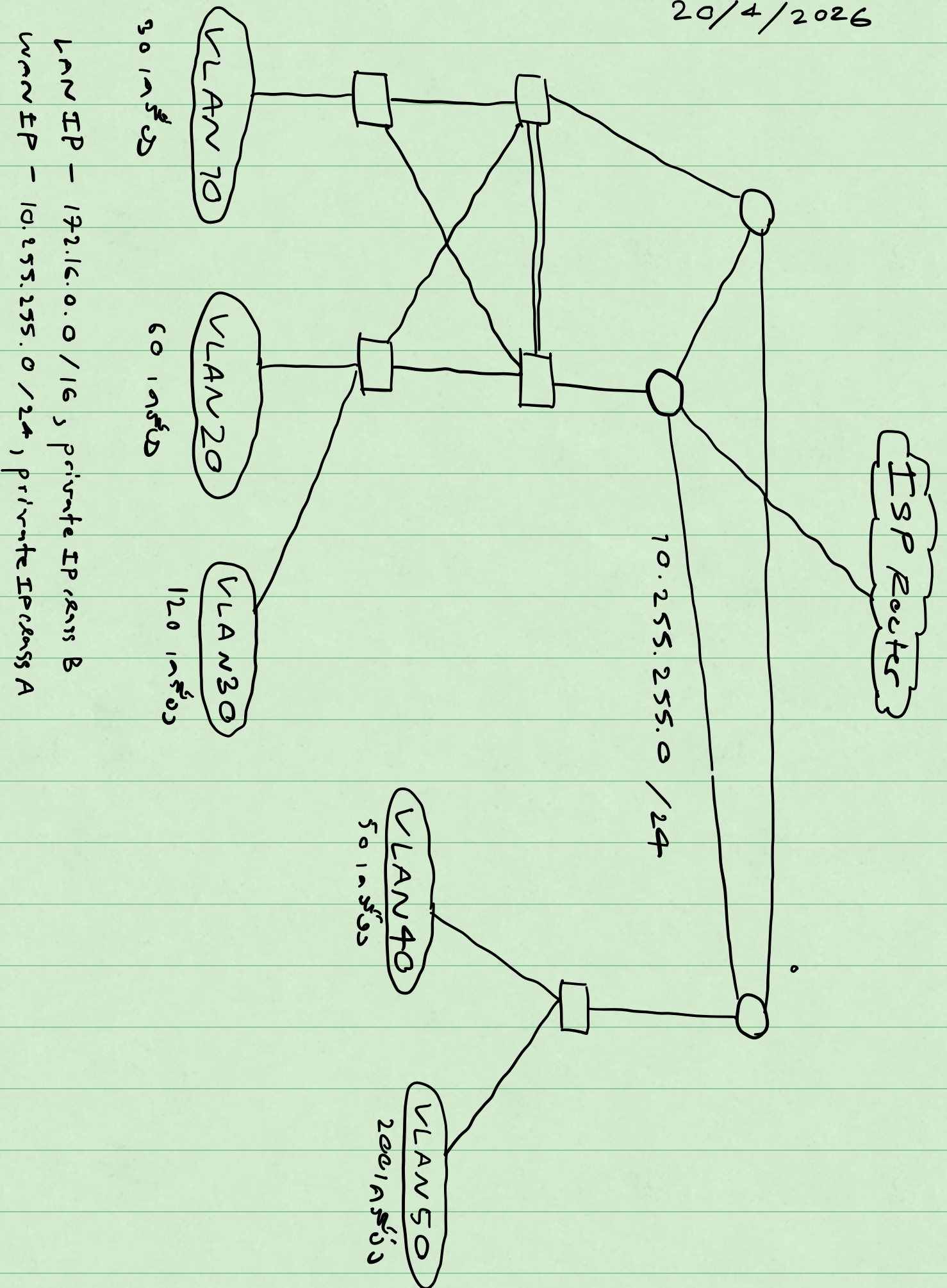
• global config

• interface config

○ VLAN

End of day

20/4/2026



Q16716 IP requirements VLAN 11 = IP address WAN

1) 11 LAN

172.16.0.0/16 as requirement of

VLAN 70 30 hosts 5

VLAN 20 60 11 3

VLAN 30 120 11 2

VLAN 40 50 11 4

VLAN 50 200 11 7 11 hosts 70² VLSM 32
8
24

subnet VLAN 50

70² 200 hosts 11 hosts 2⁸ - 2 = 254 hosts 8 bits

network address 172.16.0.0/24

subnet octet 3 11 = 4

00000000 | 00000000
1 | 00000000

next network address of 172.16.1.0/24

11 hosts broadcast address of 172.16.0.255/24

o.o VLAN 50 network @ 172.16.0.0/24

broadcast @ 172.16.0.255/24

first usable @ 172.16.0.1/24

last usable @ 172.16.0.254/24

သို့သော်လည်းကောင်း VLAN 30

32-
27
25

၇၂၂ ၁၂၀ ခုအတွက် ၁၁၂၀၀၀၀၀ ၂^၇-၂=၁၂၆ ခုအတွက် ၁၁၂၀၀

network address အဖြစ် 172.16.1.0/25

သို့သော်လည်းကောင်း octet 4

0|0000000

next subnet @ = 128

1|0000000

၁၁၂၀၀၀၀ broadcast address အဖြစ် 172.16.1.127/25

∴ VLAN 30 network @ 172.16.1.0/25

broadcast @ 172.16.1.127/25

first usable @ 172.16.1.1/25

last usable @ 172.16.1.126/25

သို့သော်လည်းကောင်း VLAN 20

၇၂၂ ၆၀ ခုအတွက် ၁၁၂၀၀၀၀၀ ၂^၆-၂=၆၂ ခုအတွက် ၁၁၂၀၀

network address အဖြစ် 172.16.1.128/26

သို့သော်လည်းကောင်း octet 4

10|000000

next subnet @ = 192

11|000000

၁၁၂၀၀၀၀ broadcast address အဖြစ် 172.16.1.191/26

∴ VLAN 20 network @ 172.16.1.128/26

broadcast @ 172.16.1.191/26

first usable @ 172.16.1.129/26

last usable @ 172.16.1.190/26

သို့သော် VLAN 40

၇၀၂ ၅၀ ကိန်းသေ ကိန်းသေ $2^6 - 2 = 62$ ကိန်းသေ ကိန်းသေ

network address ကို 172.16.1.192 /26

သို့သော် octet 3,4 00000001 11|0000000

next subnet address = .2.0 00000002 00|0000000

ကိန်းသေ broadcast address ကို 172.16.1.255

• VLAN 40 network @ 172.16.1.192 /26

broadcast @ 172.16.1.255 /26

first usable @ 172.16.1.193 /26

last usable @ 172.16.1.254 /26

သို့သော် VLAN 10

၇၀၂ ၃၀ ကိန်းသေ ကိန်းသေ $2^5 - 2 = 30$ ကိန်းသေ ကိန်းသေ

network address ကို 172.16.2.0 /27

သို့သော် octet 4 000|00000

next subnet address = .32 001|00000

ကိန်းသေ broadcast address ကို 172.16.2.31 /27

• VLAN 10 network @ 172.16.2.0 /27

broadcast @ 172.16.2.31 /27

first usable @ 172.16.2.1 /27

last usable @ 172.16.2.30 /27



2) 11000 WAN structure router 3 WAN

10.255.255.0/24

WAN 1

using 2 routers means $2^2 - 2 = 2$ subnets

octet 4

000000|00

next subnet @ = .4

000001|00

∴ router 1 = 10.255.255.1

" 2 = 10.255.255.2

WAN 2

unique network @ 10.255.255.4/30

∴ router 1 = 10.255.255.5/30

router 2 = 10.255.255.6/30

broadcast @ 10.255.255.7/30

WAN 3

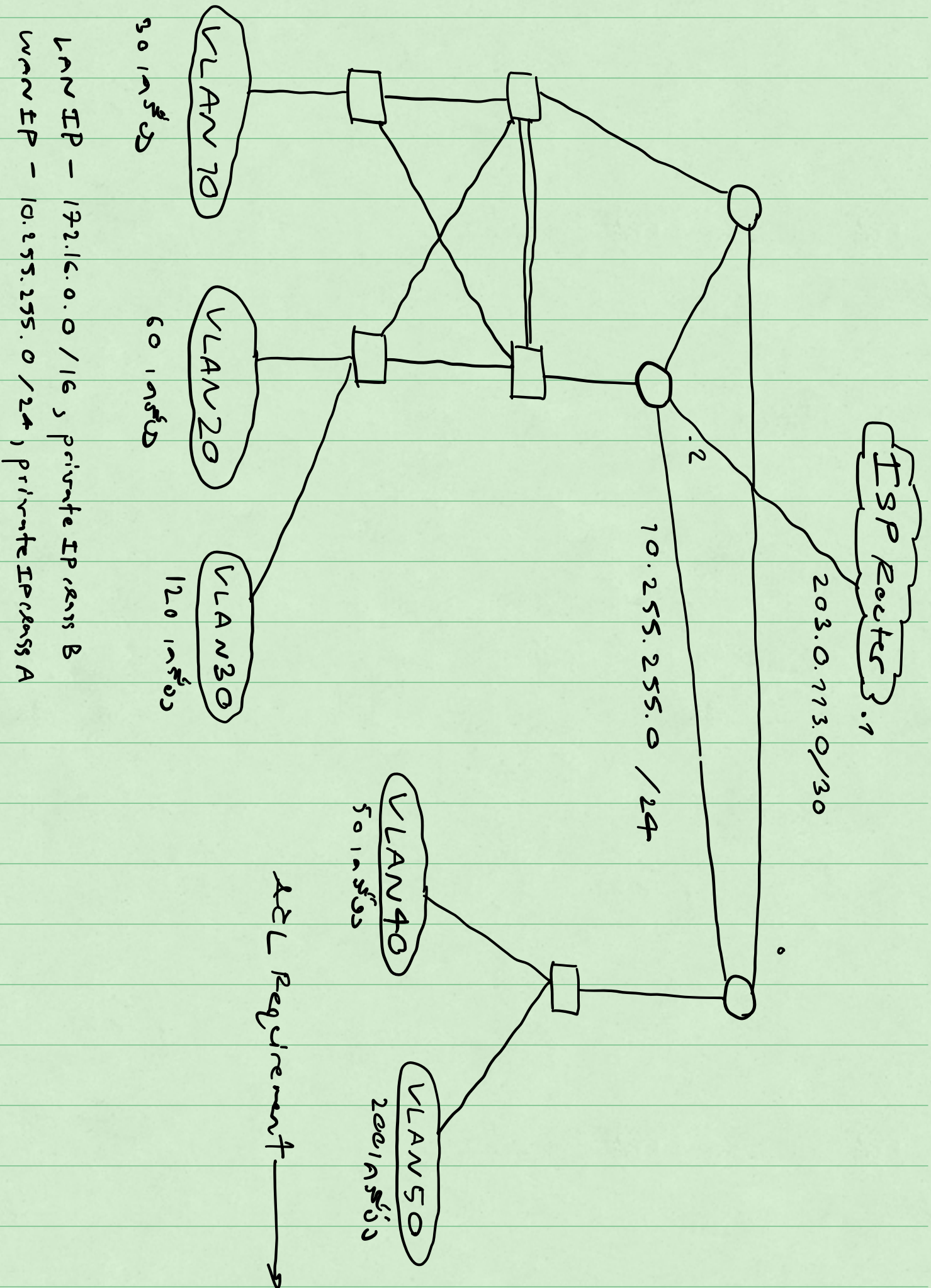
unique network @ 10.255.255.8/30

∴ router 1 = 10.255.255.9/30

router 2 = 10.255.255.10/30

broadcast @ 10.255.255.11/30

23/04/2026



VLAN 10: IT VLAN 20: Finance VLAN 30: HQ Op.

VLAN 40: BR Op. VLAN 50: Guest

ACLs policies

- 1) ηη VLAN κέρνουν VLAN 10
- 2) κέρνουν VLAN 10 VLAN 20 και VLAN 10 (IT)
- 3) κέρνουν VLAN 50 VLAN 30, VLAN 40